

|                                                           |                                              |                                      |                                                  |                                        |
|-----------------------------------------------------------|----------------------------------------------|--------------------------------------|--------------------------------------------------|----------------------------------------|
| This is sample data provided<br>for CALIBER purposes only |                                              | ONE POINT LESSON (OPL)               |                                                  | OPL No: OPL-FA-8901-05<br>Sheet 1 of 1 |
| OPL Title                                                 | Reflux Pump Low-Level Protection (LSLL-8901) | Discipline                           | Instrument                                       |                                        |
| Equipment                                                 | FA-8901 - REFLUX ACCUMULATOR DRUM            | Area / Unit                          | 8900 - HEXANE COLUMN OVERHEAD SYSTEM             |                                        |
| Related Interlock                                         | SEQ-8901 (SIL 1)                             | P&ID; Ref                            | TJC-LLD-PID-8901                                 |                                        |
| Classification:                                           | <input type="checkbox"/> Basic Knowledge     | <input type="checkbox"/> Improvement | <input checked="" type="checkbox"/> Trouble Case | Aspect: P-Q-C-D-S-M-E                  |

## 1. PURPOSE / OBJECTIVE

This One Point Lesson explains **reflux pump low-level protection (lsll-8901)** for FA-8901 (REFLUX ACCUMULATOR DRUM) in the HEXANE COLUMN OVERHEAD SYSTEM. A low drum level cavitates and damages the GA-8920 reflux pump; the LSLL trip must be proven and never casually bypassed. The lesson gives operators and technicians the key knowledge and the correct step-by-step method so the task is done safely, correctly and consistently every time.

## 2. SAFETY PRECAUTIONS

Hazard note: this equipment is **HIGH CRITICAL** handling **Hexane / water** at up to 10 barg / FV / 120 degC. Observe the following before and during the task:

- De-energise and isolate the loop; place the function to bypass/inhibit under permit.
- Inform the panel operator before working on any trip, valve or transmitter.
- Take care with instrument air / hydraulic pressure when disconnecting tubing.
- Verify intrinsic-safety / Ex integrity is restored before returning to service.
- Wear correct PPE and follow ESD/override control procedures.

## 3. TOOLS & MATERIALS REQUIRED

- HART communicator / positioner software (e.g. ValveLink)
- Loop calibrator & decade box / pressure source
- Digital multimeter and clamp meter
- Span/zero test gas or reference standards
- Torque screwdriver and small hand tools
- Certified test/inhibit permit

## 4. DETAILED PROCEDURE / STEPS

| Step | Action                                             | Check / Acceptance             |
|------|----------------------------------------------------|--------------------------------|
| 1    | Confirm the LSLL-8901 trip stops GA-8920           | Loop isolated / under permit   |
| 2    | Test the trip during commissioning and proof tests | Reference standard traceable   |
| 3    | Never bypass the trip without an authorised permit | As-found & as-left recorded    |
| 4    | Investigate any repeated low-level events          | Trip / control action verified |
| 5    | Verify the pump restarts only after level recovers | Return to service confirmed    |

## 5. COMMON PROBLEMS & TROUBLESHOOTING

| Symptom                                         | Likely Cause                              | Action                                                     |
|-------------------------------------------------|-------------------------------------------|------------------------------------------------------------|
| Reflux drum level transmitter drifting vs gauge | Transmitter zero drift                    | Recalibrated LT-8901 against the bridge gauge              |
| Blanket / vent split-range valve hunting        | Split-range calibration overlap incorrect | Re-tuned PCV-8905 split-range, verified PT-8902 hold       |
| Proof test of drum high-high level trip         | Within tolerance                          | Recalibrated LSHH-8901 and verified EFF-1 column feed trip |

## 6. KEY LEARNING POINTS

- A low drum level cavitates and damages the GA-8920 reflux pump; the LSLL trip must be proven and never casually bypassed.
- Respect the interlock SEQ-8901: e.g. LSHH-8901 trips at > 85 % - never defeat it without a permit.
- Always cross-check the Datasheet limits and the P&ID; TJC-LLD-PID-8901 before working on FA-8901.
- Record the work in the equipment history so the Knowledge Hub stays the single source of truth.

| Prepared by                 | Reviewed by (Supervisor) | Approved by (Manager)       | Date of Sharing       |
|-----------------------------|--------------------------|-----------------------------|-----------------------|
| Panel Operator / Technician | Yudha Permana (EMP-1124) | Zulkarnain Hasan (EMP-0901) | Tuesday, 23 June 2026 |